

DDL SCRIPTS

1. Trader Table

```
CREATE TABLE Trader (  
    Trader_ID INT PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Email VARCHAR(150) NOT NULL UNIQUE,  
    Balance DECIMAL(15,2) NOT NULL DEFAULT 0.00  
);
```

2. Stock_Symbol Table

```
CREATE TABLE Stock_Symbol (  
    Symbol VARCHAR(10) PRIMARY KEY,  
    Company_Name VARCHAR(200) NOT NULL  
);
```

3. Stock_Base Table

```
CREATE TABLE Stock_Base (  
    Stock_ID INT PRIMARY KEY,  
    Symbol VARCHAR(10) NOT NULL,  
    FOREIGN KEY (Symbol) REFERENCES  
    Stock_Symbol(Symbol)  
);
```

4. Administrator Table

```
CREATE TABLE Administrator (  
    Admin_ID INT PRIMARY KEY,  
    Email VARCHAR(150) NOT NULL UNIQUE,
```

```
Password_Hash VARCHAR(255) NOT NULL,  
Role_Level INT DEFAULT 1,  
Last_Login TIMESTAMP  
);
```

5. Trade Table

```
CREATE TABLE Trade (  
    Trade_ID INT PRIMARY KEY,  
    Execution_Price DECIMAL(15,2) NOT NULL,  
    Quantity INT NOT NULL,  
    Trade_Time TIMESTAMP NOT NULL  
);
```

6. Order Table

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY,  
    trader_id INT NOT NULL,  
    stock_id INT NOT NULL,  
    quantity INT NOT NULL,  
    limit_price DECIMAL(15,2) NOT NULL,  
    status VARCHAR(20) NOT NULL,  
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    order_type VARCHAR(20) NOT NULL,  
    FOREIGN KEY (trader_id) REFERENCES trader(trader_id),  
    FOREIGN KEY (stock_id) REFERENCES  
stock_base(stock_id)  
);
```

7. Portfolio Table

```
CREATE TABLE Portfolio (  
    Trader_ID INT,  
    Stock_ID INT,  
    Quantity INT DEFAULT 0,  
    PRIMARY KEY (Trader_ID, Stock_ID),  
    FOREIGN KEY (Trader_ID) REFERENCES  
Trader(Trader_ID),  
    FOREIGN KEY (Stock_ID) REFERENCES  
Stock_Base(Stock_ID)  
);
```

8. Trade_Order Table

```
CREATE TABLE trade_order (  
    trade_id INT,  
    order_id INT,  
    PRIMARY KEY (trade_id, order_id),  
    FOREIGN KEY (trade_id) REFERENCES trade(trade_id),  
    FOREIGN KEY (order_id) REFERENCES orders(order_id)  
);
```

9. System_Audit_Log Table

```
CREATE TABLE system_audit_log (  
    log_id SERIAL PRIMARY KEY,  
    actor_id INT NOT NULL,  
    action_type VARCHAR(50) NOT NULL,  
    action_timestamp TIMESTAMP DEFAULT  
CURRENT_TIMESTAMP,  
    target_entity_id INT  
);
```

INSERTION QUERIES

1. INSERT INTO Trader

```
INSERT INTO Trader VALUES  
(101, 'Aaditya Dave', 'aaditya1152@gmail.com', 500000.00),  
(102, 'Janvi Patel', 'janvi2611@gmail.com', 750000.00),  
(103, 'Femina Rathod', 'femina200611@gmail.com', 300000.00),  
(104, 'Nirva Nadpara', 'nirva2006@gmail.com', 450000.00),  
(105, 'Karan Desai', 'karan@gmail.com', 600000.00);
```

2. INSERT INTO Stock_Symbol

```
INSERT INTO Stock_Symbol VALUES  
( 'RELIANCE', 'Reliance Industries'),  
( 'TCS', 'Tata Consultancy Services'),  
( 'INFY', 'Infosys Ltd'),  
( 'HDFCBANK', 'HDFC Bank'),  
( 'ICICIBANK', 'ICICI Bank');
```

3. INSERT INTO Stock_Base

```
INSERT INTO Stock_Base VALUES  
(1, 'RELIANCE'),  
(2, 'TCS'),  
(3, 'INFY'),  
(4, 'HDFCBANK'),  
(5, 'ICICIBANK');
```

4. INSERT INTO Administrator

```
INSERT INTO Administrator VALUES  
(1, 'admin1@stock.com', 'hash123', 1, NOW()),  
(2, 'admin2@stock.com', 'hash456', 2, NOW());
```

5. INSERT INTO Portfolio

```
INSERT INTO Portfolio VALUES  
(101, 1, 50),  
(101, 2, 30),  
(102, 3, 100),  
(103, 1, 20),  
(104, 4, 70),  
(105, 5, 40);
```

6. INSERT INTO Order

```
INSERT INTO orders VALUES  
(201, 101, 1, 10, 2500.00, 'PENDING', NOW(), 'BUY'),  
(202, 102, 2, 5, 3500.00, 'OPEN', NOW(), 'SELL'),  
(203, 103, 3, 20, 1500.00, 'FILLED', NOW(), 'BUY'),  
(204, 104, 4, 15, 1600.00, 'CANCELLED', NOW(), 'SELL'),  
(205, 105, 5, 25, 900.00, 'PENDING', NOW(), 'BUY'),  
(206, 101, 2, 12, 3400.00, 'FILLED', NOW(), 'SELL');
```

7. INSERT INTO Trade

```
INSERT INTO Trade VALUES  
(301, 2480.00, 10, NOW()),  
(302, 3450.00, 5, NOW()),  
(303, 1490.00, 20, NOW()),  
(304, 1580.00, 15, NOW()),
```

```
(305, 910.00, 25, NOW());
```

8. INSERT INTO Trade_Order (Bridge Table)

```
INSERT INTO Trade_Order VALUES  
(301, 201),  
(302, 202),  
(303, 203),  
(304, 204),  
(305, 205);
```

9. INSERT INTO System_Audit_Log

```
INSERT INTO System_Audit_Log (Actor_ID, Action_Type,  
Target_Entity_ID) VALUES  
(1, 'ADD_STOCK', 1),  
(2, 'DELETE_ORDER', 204),  
(1, 'UPDATE_PRICE', 2),  
(2, 'ADD_TRADER', 105);
```